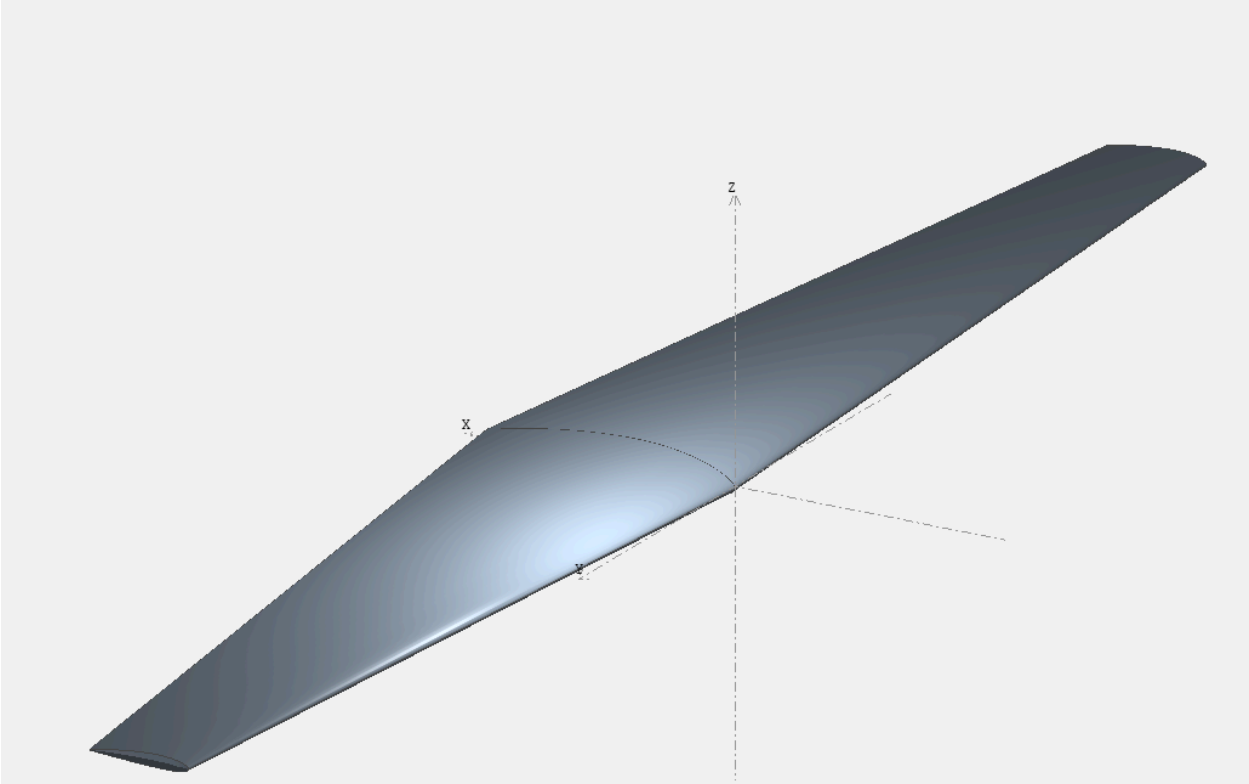
Wing Design Project

Group Number 1
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Course: EAS 4101
Date: April 23, 2026




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Plane Name
Wing Span      =     9.000 m
xyProj. Span   =     8.999 m
Wing Area      =     8.001 m²
xyProj. Area   =     8.000 m²
Plane Mass     =     0.000 kg
Wing Load      =     0.000 kg/m²
Root Chord     =     1.270 m
MAC            =     0.943 m
TipTwist       =     0.000°
Aspect Ratio   =    10.124
Taper Ratio    =     0.400
Root-Tip Sweep =     0.000°
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Design Discussion

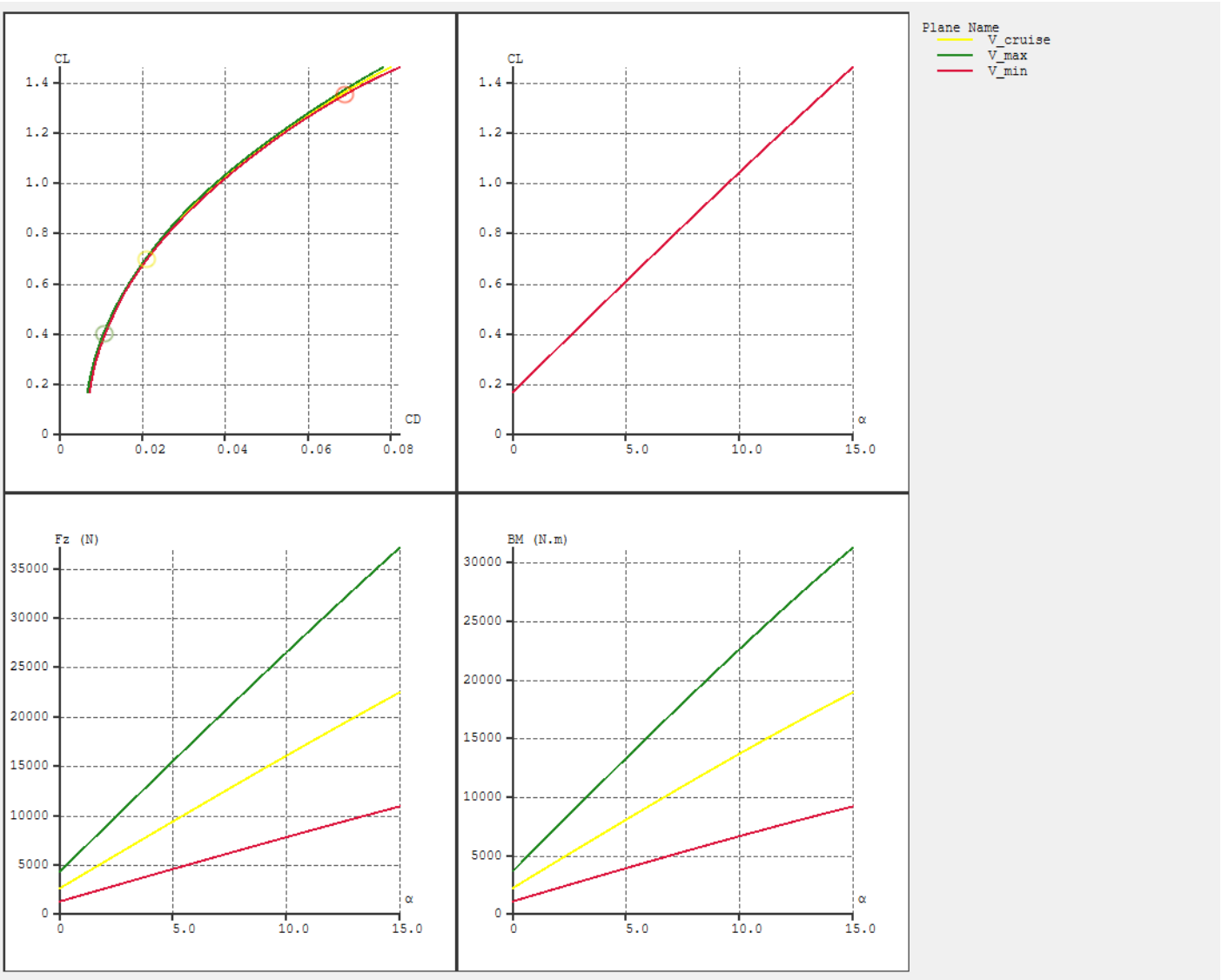
To determine the design specifications for the wing, we first found that the NACA 4412 and 2412 airfoils were excellent options for efficiency from online resources. The decision to place the NACA 2412 airfoil at the tip of the wing was based on its lower camber, reducing the drag profile at the tip and lessening the bending moment compared to the 4% cambered NACA 4412. In order to maintain steady, level flight, the lift generated must equal the total weight of the aircraft, *L = W = 10,231.8 N*. To determine the planform area, we used the equation $S = \frac{L}{\frac{1}{2}\rho V_{min}^2 \cdot C_{L,max}}$. Plugging in a maximum lift coefficient of 1.4, a typical value for our selected airfoils, the equation yielded: $S \approx 8m^2$. Continuing on, a taper ratio of 0.4 was chosen to achieve a near elliptic lift distribution since the design problem calls for a high lift-to-drag ratio. From here, we determined the root and tip chords using $c_r = \frac{2S}{b(1+\lambda)}$ and $c_t = c_r \cdot \lambda$. After determining our desired specifications, we checked the approximate root moment with $M_r \approx \frac{Wb}{12}(\frac{1+2\lambda}{1+\lambda})$, ensuring the calculated value was under the $11,000\ N \cdot m$ threshold. It was during this constraint check that we found an 11 meter wingspan to be too long given the taper ratio we decided on. To combat this, we simply lowered the wingspan to 9 meters resulting in $M_r \approx 9,866.38\ N \cdot m$, leaving plenty of room before reaching the maximum bending moment *(note: the calculated bending moment above is an ideal approximation and not identical to our analytic values)*. This change creates a safety barrier for the structure while also leaving room for a longer wingspan if the lift-to-drag ratio is too low. Next, to meet the manufacturing constraint of zero sweep at the quarter chord, we applied a spanwise offset to the tip which ensured the aerodynamic center remained unswept across the span. Finally, we set the angle of twist to -2° to create a favorable stall profile that keeps the flight control surfaces at the wing tips effective at higher angles of attack.

Table 1: Wing Parameters

Wing Span (m)	Root Chord (m)	Tip Chord (m)	Planform Area (m^2)	Aspect Ratio
9.00	1.270	0.508	8.00	10.12
Airfoil Section at Root	Airfoil Section at Tip	Dihedral Angle	Sweep Angle (At ¼ Chord)	Twist Angle
NACA 4412	NACA 2412	1.72°	0.0°	-2.00°

Table 2: Key Operating Conditions

Airspeed (m/s)	AOA (degrees)	Re (at root)	CL	dCL/dα (1/radian)	CL/CD	Lift Fz (N)	Root BM (N-m)
V_{max} = 72	2.62°	6.096e6	0.3999	4.926	38.01	10231.8	8723.0
V_{cruise} = 56	5.65°	4.741e6	0.6657	4.926	34.09	10231.8	8777.9
V_{min} = 39	13.96°	3.302e6	1.3729	4.926	19.24	10231.8	8645.5


CL
CD
CL
α
Fz (N)
α
BM (N.m)
α
Plane Name
V_cruise
V_max
V_min